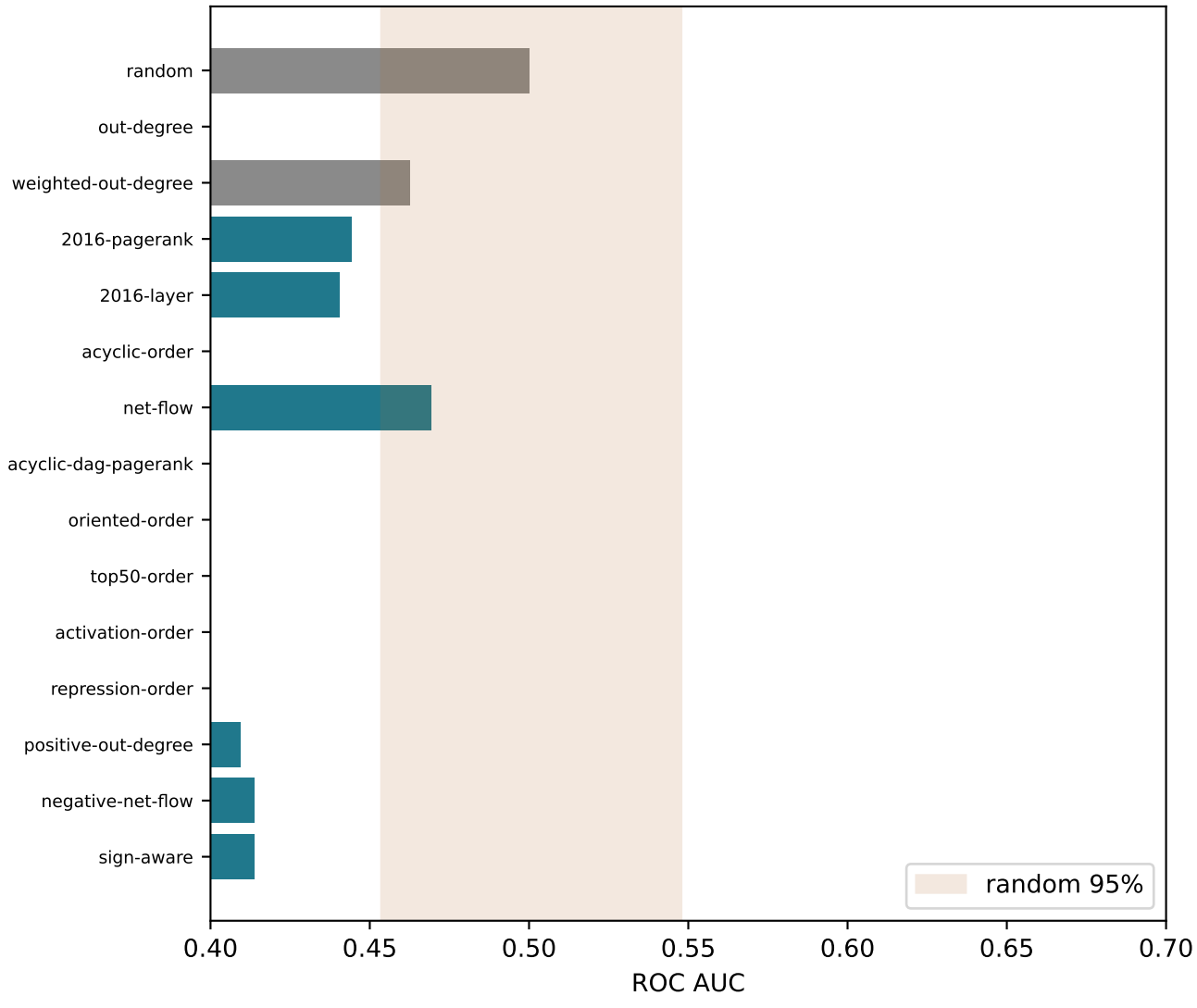
Regulator ranking against reference 'regulators'

2026-11-07 · 739 genes ranked, 199 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	26.8 (19–35)	0.500 (0.454–0.548)				
out-degree	5	0.352	1.000	1.000	-0.110 (-0.127, -0.094)	< 0.001
weighted-out-degree	6	0.463	1.000	0.938	+0.000 (+0.000, +0.000)	
2016-pagerank	29	0.444	0.334	0.991	-0.018 (-0.059, +0.029)	0.458
2016-layer	80	0.441	< 0.001	0.993	-0.022 (-0.063, +0.022)	0.296
acyclic-order	5	0.324	1.000	1.000	-0.139 (-0.169, -0.106)	< 0.001
net-flow	6	0.469	1.000	0.895	+0.007 (-0.004, +0.018)	0.236
acyclic-dag-pagerank	5	0.336	1.000	1.000	-0.127 (-0.148, -0.106)	< 0.001
oriented-order	7	0.290	1.000	1.000	-0.172 (-0.210, -0.136)	< 0.001
top50-order	27	0.383	0.520	1.000	-0.080 (-0.122, -0.037)	< 0.001
activation-order	6	0.333	1.000	1.000	-0.130 (-0.164, -0.095)	< 0.001
repression-order	7	0.380	1.000	1.000	-0.083 (-0.118, -0.045)	< 0.001
positive-out-degree †	5	0.409	1.000	1.000	-0.053 (-0.074, -0.030)	< 0.001
negative-net-flow †	6	0.414	1.000	1.000	-0.049 (-0.069, -0.028)	< 0.001
sign-aware †	5	0.414	1.000	1.000	-0.049 (-0.060, -0.038)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

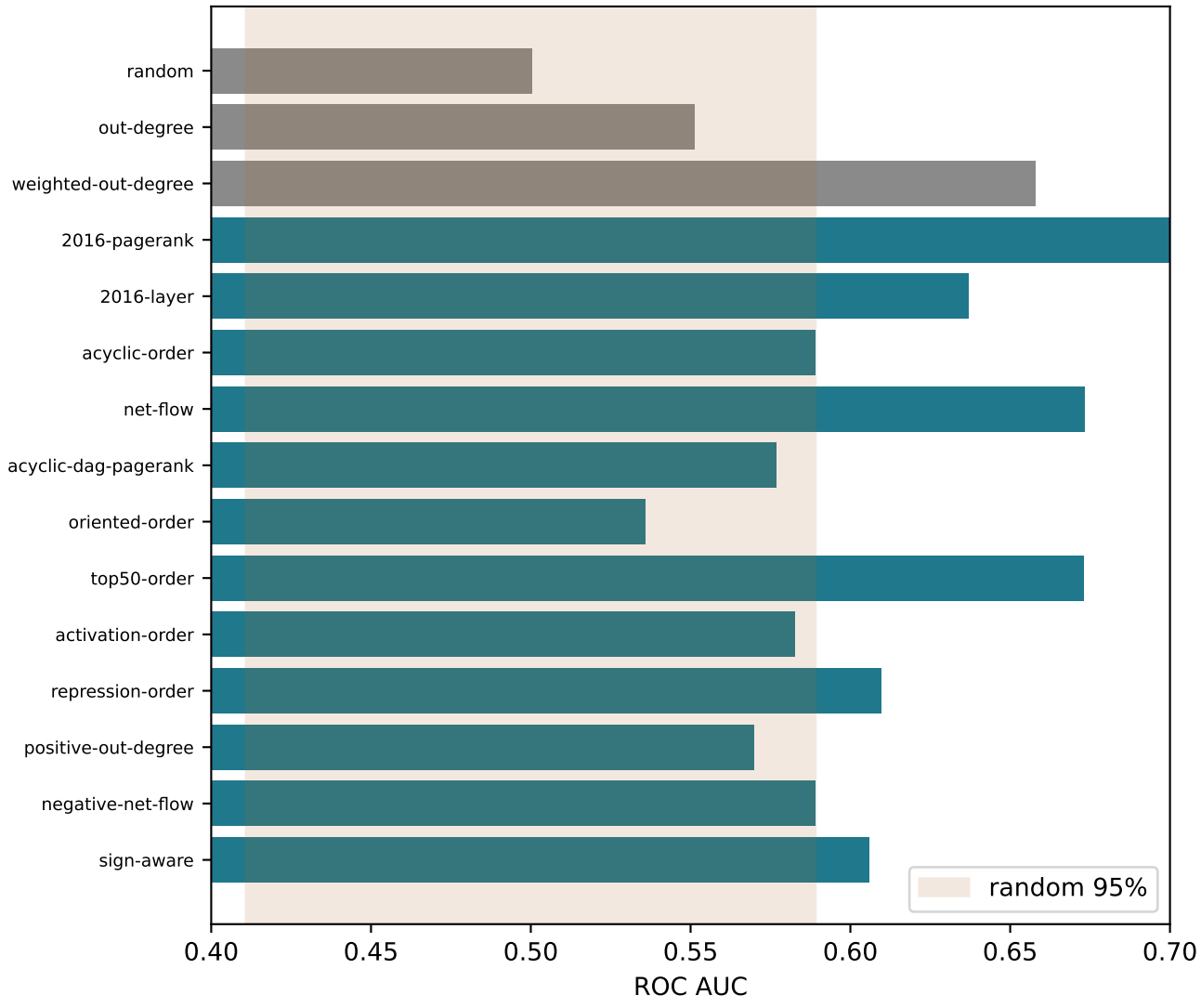


Regulator ranking against reference 'top-layer'

2026-11-07 · 739 genes ranked, 41 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	5.5 (2–10)	0.500 (0.411–0.589)				
out-degree	2	0.551	0.981	0.123	-0.107 (-0.145, -0.070)	< 0.001
weighted-out-degree	2	0.658	0.981	< 0.001	+0.000 (+0.000, +0.000)	
2016-pagerank	13	0.732	0.002	< 0.001	+0.074 (+0.014, +0.137)	0.010
2016-layer	30	0.637	< 0.001	0.003	-0.021 (-0.091, +0.050)	0.546
acyclic-order	2	0.589	0.981	0.025	-0.069 (-0.130, -0.014)	0.010
net-flow	2	0.673	0.981	< 0.001	+0.015 (-0.001, +0.032)	0.062
acyclic-dag-pagerank	2	0.577	0.981	0.047	-0.081 (-0.123, -0.047)	< 0.001
oriented-order	5	0.536	0.664	0.215	-0.122 (-0.207, -0.053)	0.004
top50-order	13	0.673	0.002	< 0.001	+0.015 (-0.068, +0.089)	0.660
activation-order	2	0.583	0.981	0.033	-0.075 (-0.136, -0.021)	0.008
repression-order	4	0.610	0.830	0.012	-0.048 (-0.114, +0.020)	0.172
positive-out-degree †	2	0.570	0.981	0.064	-0.088 (-0.132, -0.043)	< 0.001
negative-net-flow †	4	0.589	0.830	0.026	-0.069 (-0.104, -0.037)	< 0.001
sign-aware †	2	0.606	0.981	0.014	-0.052 (-0.081, -0.027)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	3,561	0.800 (42.61×)	50.30×	31.63×	1.00 (0.58)
resolve_reciprocals → drop_sinks → prune_triangles	2,365	0.119 (33.18×)	39.12×	27.15×	1.00 (0.52)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	1,002	0.126 (41.68×)	42.68×	40.76×	1.00 (0.50)
absolute	3,561	0.800 (42.61×)			
absolute → minimum_feedback_order	3,561	0.596 (39.14×)			
orient_reciprocals ratio=2.0 → absolute	3,561	0.514 (38.00×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	3,561	0.514 (38.00×)			
top_outgoing k=50 → absolute	3,561	0.800 (49.24×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,561	0.601 (45.54×)			
keep_sign sign=1 → absolute	3,561	0.556 (50.30×)			
keep_sign sign=1 → absolute → minimum_feedback_order	3,561	0.419 (47.21×)			
keep_sign sign=-1 → absolute	3,561	0.245 (31.63×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	3,547	0.186 (28.64×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 40,827 edges (19,397 negative), largest strong component 735.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 40,827 edges (19,397 negative), largest strong component 735.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,721 edges (4,168 negative), largest strong component 104.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,721 edges (4,168 negative), largest strong component 104; DAG keeps 3,650 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 40,827 edges (0 negative), largest strong component 735; DAG keeps 36,209 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 40,827 edges (0 negative), largest strong component 735.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 40,827 edges (0 negative), largest strong component 735; DAG keeps 36,209 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 31,806 edges (0 negative), largest strong component 1; DAG keeps 31,806 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 26,954 edges (0 negative), largest strong component 735; DAG keeps 23,211 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 21,430 edges (0 negative), largest strong component 727; DAG keeps 18,865 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 19,397 edges (0 negative), largest strong component 710; DAG keeps 17,351 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 40,827 edges (19,397 negative), largest strong component 735.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 40,827 edges (19,397 negative), largest strong component 735.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 40,827 edges (19,397 negative), largest strong component 735.